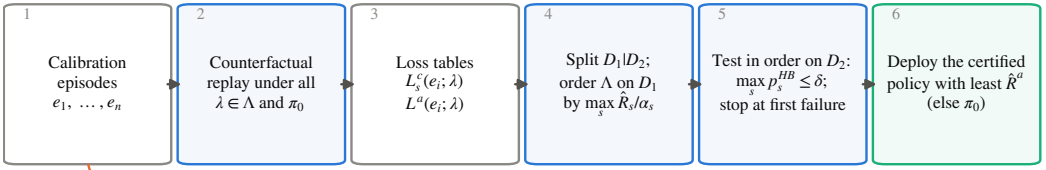
1
Calibration episodes $e_1, \ldots, e_n$
2
Counterfactual replay under all $\lambda \in \Lambda$ and π_0
3
Loss tables $L_s^c(e_i;\lambda)$ $L^a(e_i;\lambda)$
4
Split $D_1|D_2$; order Λ on D_1 by $\max_s \hat{R}_s/\alpha_s$
5
Test in order on D_2: $\max_s p_s^{HB} \le \delta$; stop at first failure
6
Deploy the certified policy with least $\hat{R}^a$ (else π_0)
Slice-FPR baseline (Corollary 1)
benign scores only, no replay;
test $\Phi_s(\tau) \le \alpha_s$ per slice at $\delta/3$
Guarantee (Proposition 1)
$\Pr\{\forall s : R_s^c(\hat{\lambda}) \le \alpha_s\} \ge 1-\delta$
no monotonicity assumption